PV101 Day 8 Student Worksheet Answer Key

Instructor copy

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Formula and units check

- Power formula: W = V x A.
- 24 W x 3 h = 72 Wh.
- 0.5 kW x 1,000 W/kW = 500 W.
- In series, voltage adds. In matched parallel paths, current adds.

Verification chain

Symptom -> test -> evidence -> conclusion. Accept equivalent wording that keeps the observation before the test and the conclusion after evidence.

System path

Accept PV -> controller -> battery -> protected inverter/DC load path with a complete return, one source-side fuse, one disconnect, and one logical meter point.

Misconceptions

Statement	Answer	Correction
Wire color proves polarity.	Incorrect	Color is a clue; verify polarity with labels, tracing, and the correct meter setup.
Continuity is tested on an energized circuit.	Incorrect	Continuity/ohms testing requires the circuit to be de-energized.
Replacing a fuse proves the fault is repaired.	Incorrect	Find the cause and verify the restored known-good state.
Loads are the starting point for sizing.	Correct	Document watts, hours, simultaneous loads, and assumptions before sizing.
Cleanup is part of technical work.	Correct	A professional closeout includes safe state, material count, records, and room reset.